Module 12: Structures

12.1 Introduction to Structures

- A structure groups multiple variables of different types under one name
- Useful for representing a real-world entity that has multiple attributes
    - A student has a name, ID, age, and GPA
    - A point has x and y coordinates
    - A date has day, month, and year

12.2 Defining and Using Structures

- Define a structure using the struct keyword

```
struct Student {
string name;
int id;
double gpa;
};
```

- Declaring a structure variable

```
Student s1;
Student s1 = {"Ali", 101, 3.8};
```

- Accessing members using the dot operator

```
cout << s1.name;
s1.gpa = 3.9;
```

- Array of structures: useful for storing multiple records

```
Student roster[30]; // array of 30 Student structures
```

12.3 Structures and Functions

- Pass by value: a copy of the entire structure is made (can be slow for large structures)
- Pass by reference: use & to avoid copying

```
void printStudent(const Student &s) {
cout << s.name << " - " << s.gpa;
}
```

- Return a structure from a function

```
Student createStudent(string n, int i, double g) {
Student s;
s.name = n; s.id = i; s.gpa = g;
return s;
}
```

12.4 Pointers to Structures

- The arrow operator (->) is used to access structure members through a pointer

```
Student *ptr = &s1;
cout << ptr->name; // same as (*ptr).name
```